Collaborative Agents

Agent Architecture using LangGraph, RAG & Generative UI

Problem Statement

TRADITIONAL AI ASSISTANTS ARE MOSTLY STATELESS

LACK OF LONG-TERM AND EPISODIC MEMORY

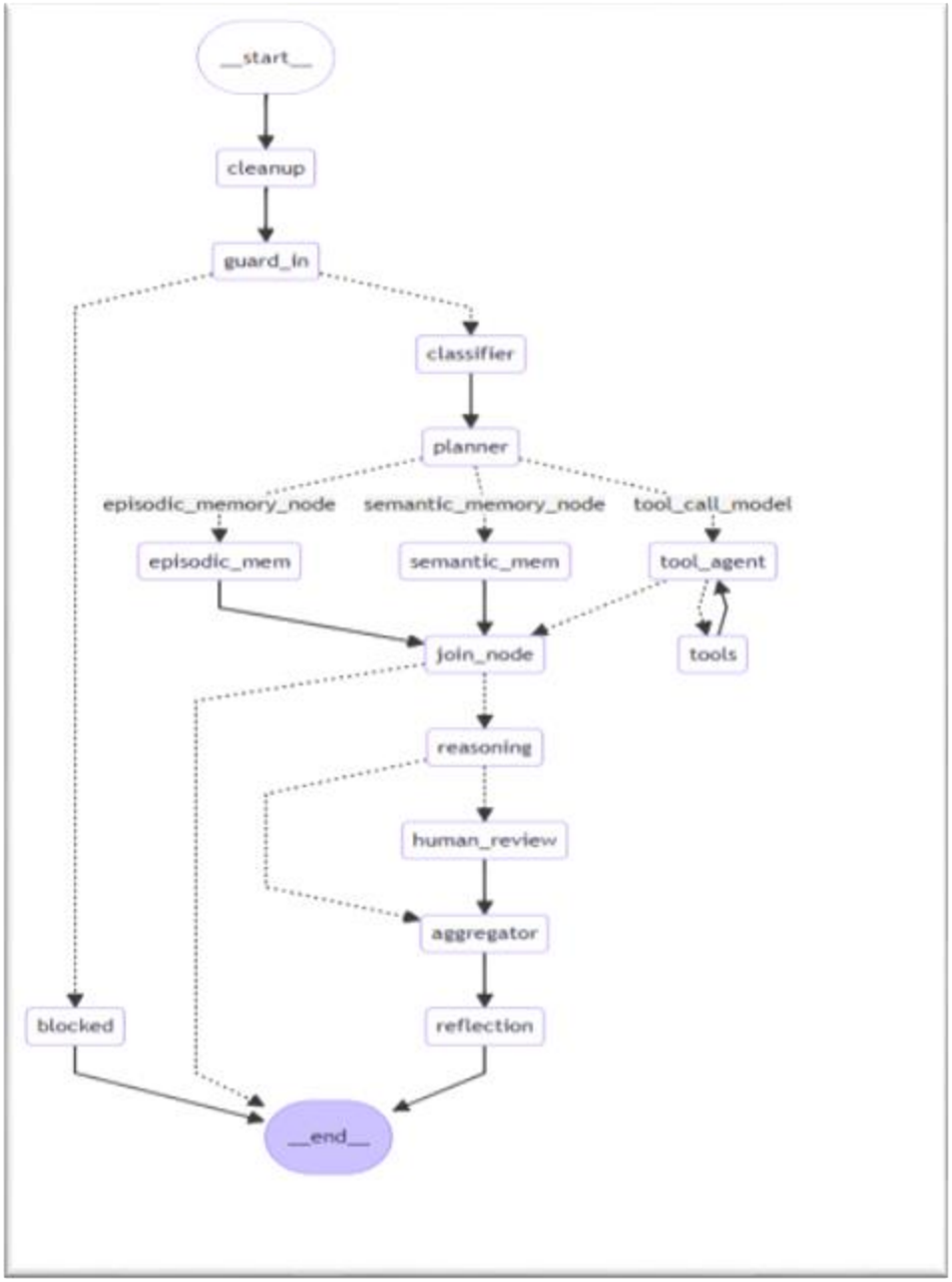
LIMITED SAFETY CONTROLS

NO HUMAN VALIDATION FOR LOW-CONFIDENCE RESPONSES

Solution Overview

- Collaborative agent system using LangGraph
- Multiple memory layers for better context
- Human-in-the-loop based on confidence
- Real-time agent state streaming to UI

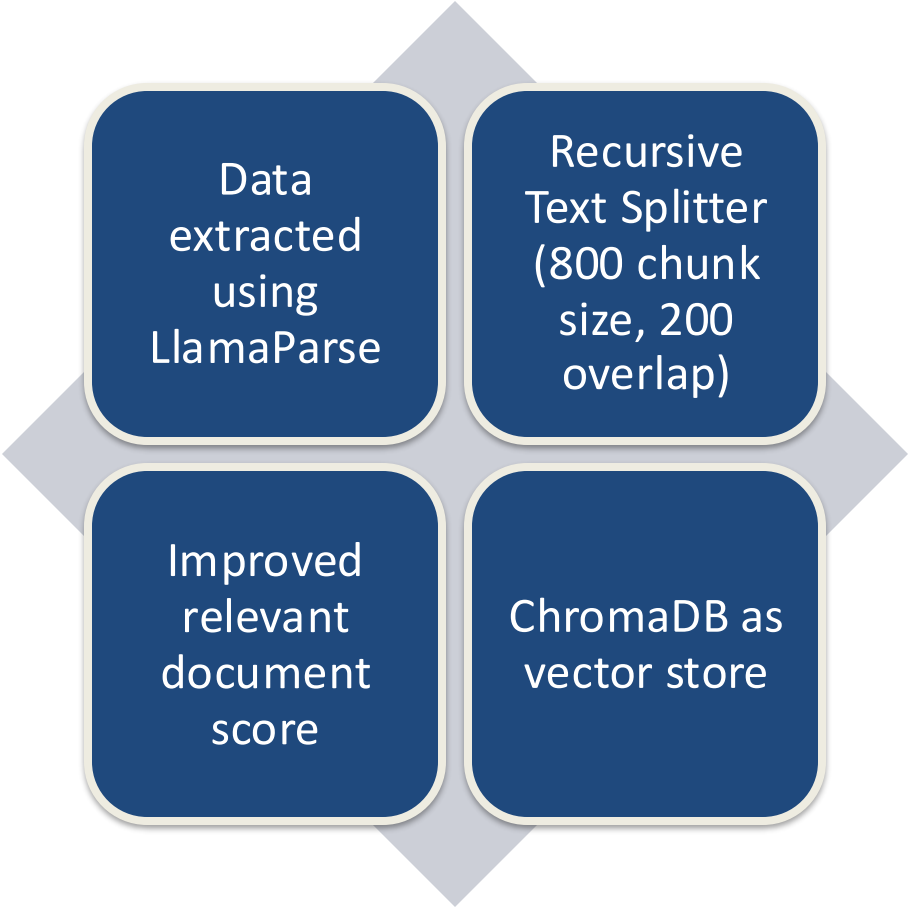
Architecture



Agent Orchestration – LangGraph

- State-driven execution model
- Supports multi-step reasoning
- Checkpointing for recovery
- Human-in-loop flow support

Long-Term Memory – RAG



Memory

SEMANTIC CONTEXT Copy

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```
- **Incident Ticket:** INC-165569
- **Severity:** P2 - High Priority
- **Date:** 2026-01-22 07:03:09 UTC
- **Investigation Team:** Site Reliability Engineering

**Executive Summary**

This document provides a comprehensive analysis of the performance
degradation experienced by the Recommendation Engine service. The
```

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```
**Performance Degradation Investigation Report**

**Recommendation Engine - High Latency Analysis**

- **Incident Ticket:** INC-165569
- **Severity:** P2 - High Priority
- **Date:** 2026-01-22 07:03:09 UTC
- **Investigation Team:** Site Reliability Engineering
```

Mem0.ai

Memories

Pick a date range | All Time | 1d | 7d

Overview | Technology | Professional Details | Misc | Filters

Created At	Entities	Memory Content	Categories	Act
2 minutes ago	default_user	User asks for the status of incident ID INC-298593		
3 minutes ago	default_user	User says yes go ahead with all	misc	
6 minutes ago	default_user	User asks: what are documents which i just uplaoded	misc	
7 minutes ago	default_user	User requests to find incident reports about service downtime	professional_details +1	
13 minutes ago	default_user	User is asking for a list of tickets they asked about yesterday	misc	
20 minutes ago	Madhu	User requests details on Incident Ticket INC-513016 from past do...	professional_details +1	
about 1 hour ago	user_xyz	User wants an executive summary for Incident Ticket INC-165569.	professional_details	
about 1 hour ago	user_xyz	User says Incident Ticket INC-165569 is not available or may not e...	professional_details +1	

Episodic & Working Memory

Episodic memory using Mem0.ai

Stores interaction history

Working memory via LangGraph checkpointers

Enables resumability

Safety – NeMo Guardrails

Input validation using NeMo Guardrails

Blocks unsafe or invalid prompts

Ensures controlled agent behavior

MCP Tool Integration

Custom MCP tool implemented

Search, create, and update tickets

Integrated directly with agent workflow

FastMCP Integration

TOOL TRACE ANALYSIS

TOOL EXECUTION
get_ticket
COMPLETED

INPUT OUTPUT


```
{
  "ticket_id": "TKT-101"
}
```


TOOL TRACE ANALYSIS

TOOL EXECUTION
get_ticket
COMPLETED

INPUT OUTPUT


```
[
  {
    "type": "text",
    "text": "{\n  \"success\": true,\n  \"ticket\": {\n
\"ticket_id\": \"TKT-101\",\n    \"subject\": \"Login issue\",\n
\"description\": \"User cannot log in with valid credentials\",\n
\"status\": \"Open\",\n    \"priority\": \"High\"\n  }\n}",
    "id": "lc_2f5412f5-d960-42d9-a3de-d732ec0e3a6d"
  }
]
```

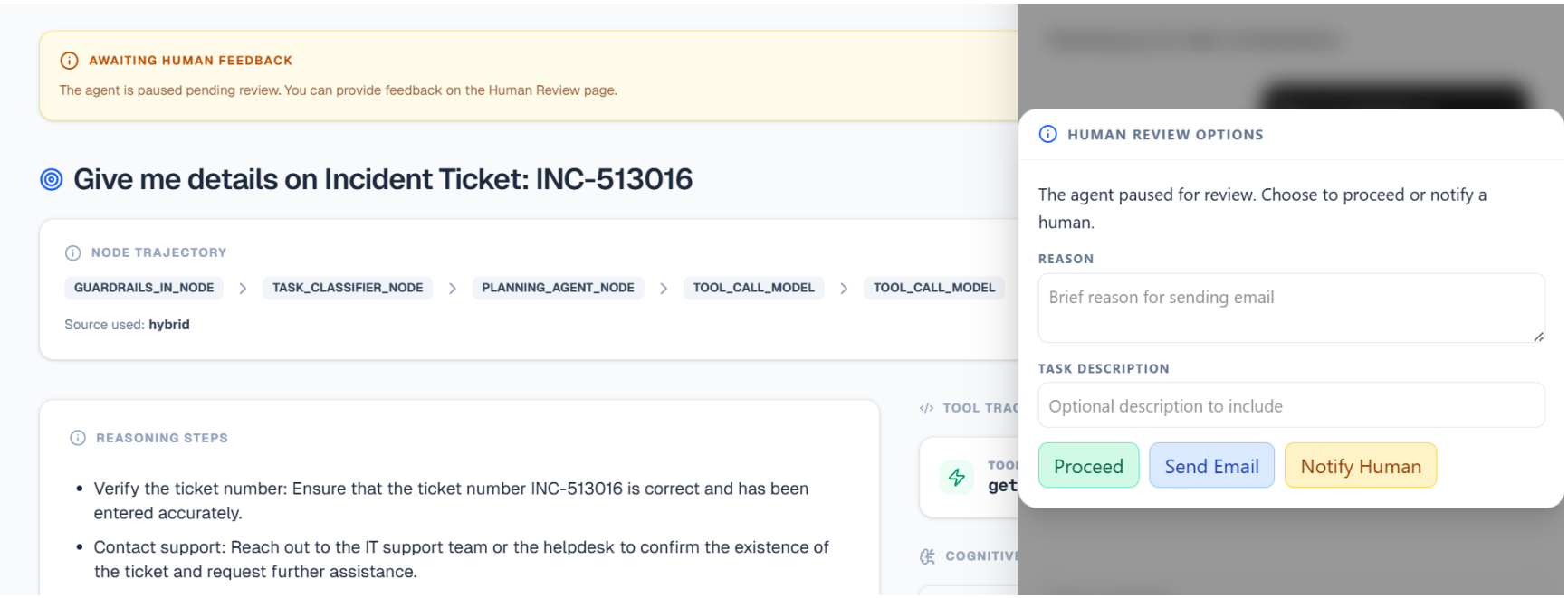
Human-in-the-Loop

AGENT CALCULATES RESPONSE CONFIDENCE

HUMAN INTERVENTION IF CONFIDENCE < 50%

USER CAN PROCEED OR NOTIFY OTHERS

Human In Loop



Frontend – Next.js & CopilotKit

Next.js based frontend

CopilotKit framework integration

Interactive AI-first UI

Generative UI – AG-UI Protocol

Streams agent state to UI

Displays reasoning and tool calls

Improves transparency

Generative UI

NEXUS // ORCHESTRATOR

UPLOAD DOCUMENTS VIEW GRAPH

NODE TRAJECTORY

GUARDRAILS_IN_NODE > TASK_CLASSIFIER_NODE > PLANNING_AGENT_NODE > SEMANTIC_AGENT_NODE > TOOL_CALL_MODEL > TOOL_CALL_MODEL > REASONING_NODE > AGGREGATOR_NODE > REFLECTION_NODE

Source used: **hybrid**

FINAL RESPONSE

- The login issue in ticket TKT-101 is currently open and marked as high priority.
- Unfortunately, the available technical documentation does not provide relevant information or resolution steps for this specific login problem.

REASONING STEPS

- Verify the user's credentials to ensure they are correct and active.
- Check if there are any ongoing issues with the authentication service that might be affecting login functionality.
- Review recent changes or updates to the login system that might have introduced this issue.
- If the issue persists, escalate to the Site Reliability Engineering team for further investigation.

EXECUTION TIMELINE

```
Classified task intent and urgency.
Planner selected: tool_calls, semantic_memory
Retrieved technical documentation
Executed tool call model.
Executed tool call model.
Synthesized reasoning from retrieved contexts.
Streamed final response to the user.
```

TOOL TRACE ANALYSIS

TOOL EXECUTION
get_ticket COMPLETED

COGNITIVE MEMORY

SEMANTIC CONTEXT Copy

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```
- **Incident Ticket:** INC-165569
- **Severity:** P2 - High Priority
- **Date:** 2026-01-22 07:03:09 UTC
- **Investigation Team:** Site Reliability Engineering

**Executive Summary**

This document provides a comprehensive analysis of the performance degradation
experienced by the Recommendation Engine service. The investigation revealed
```

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```
**Performance Degradation Investigation Report**

**Recommendation Engine - High Latency Analysis**

- **Incident Ticket:** INC-165569
- **Severity:** P2 - High Priority
- **Date:** 2026-01-22 07:03:09 UTC
- **Investigation Team:** Site Reliability Engineering
```

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```
**Performance Degradation Investigation Report**

**Data Pipeline - High Latency Analysis**
```

Generative UI

CLASSIFICATION & SAFETY

OTHER LOW LOW

Request for details on ticket TKT-101

SAFE

EXECUTION PLAN

PRIORITY STANDARD

SELECTED NODES

TOOL_CALLS SEMANTIC_MEMORY

STRATEGY NOTES

The user is requesting specific details about a ticket, which requires accessing the tool_calls node to retrieve ticket information. Additionally, semantic_memory is activated to provide any general documentation or details related to the ticket.

INTERNAL STEPS

1. Classified task intent and urgency.
2. Planner selected: tool_calls, semantic_memory
3. Retrieved technical documentation
4. Executed tool call model.
5. Executed tool call model.
6. Synthesized reasoning from retrieved contexts.
7. Streamed final response to the user.

CONFIDENCE & SOURCE

CONFIDENCE 85%

SOURCE hybrid

The request is for details on ticket TKT-101, which is related to a login issue where a user cannot log in with valid credentials. The ticket is currently open and has a high priority status. There is no direct link between this ticket and the incidents documented in the technical documentation provided. The incidents in the documentation are related to performance degradation in the Recommendation Engine and Data Pipeline services, which are unrelated to login issues. Therefore, the technical documentation does not provide additional context or resolution steps for TKT-101.

Cloud Infrastructure/DevOps Test Data

- **Package Statistics**
  Total Files- 130
  Total Pages- 547
  Total Words- 258500
  Package Memory- 3.1 MB (compressed)Format

- **File Distribution**

  30 files--- Service-Down Incidents(pdfs)
  26 files – High Latency Reports(docs)
  25 files – Deployment Failures(txts)
  15 files – Misconfigurations(Jsons)

We used Cloud Ai to generate this data

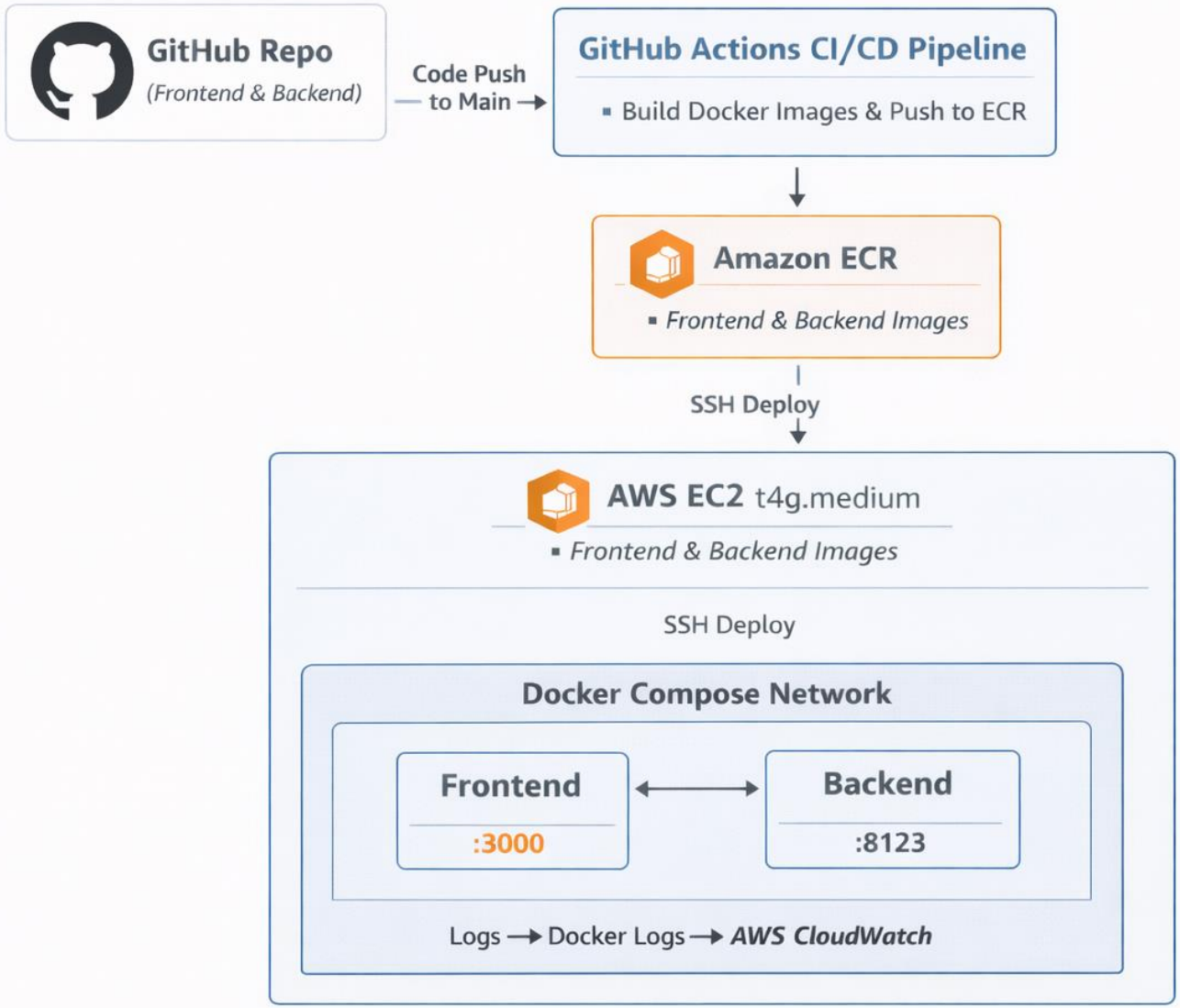
Testing Strategy

- LLM as a Judge
- Trajectory testing
- Experimental datasets
- LangSmith for evaluation and observability

Langsmith Datasets

#	Inputs	Reference Outputs	Outputs	correctness 0.70 AVG	trajectory_match 0.80 AVG	Latency 21.48 P50
1	Provide the execu...	Executive summary of inci...	- We were unable to locate the ti...	1.00	1.00	36.01s
2	Last week we dis...	Summary of prior interacti...	- I understand your concern rega...	0.00	1.00	20.21s
3	The ticket from y...	Actions and status	- I understand you're trying to cl...	1.00	1.00	28.23s
4	What is the status...	Ticket status details	- I understand you're looking for ...	1.00	1.00	19.93s
5	How do I reset m...	Password reset steps	To reset your password, please f...	1.00	1.00	22.12s
6	I'm seeing intermi...	Guidance with potential es...	I'm sorry to hear you're experien...	1.00	0.00	30.27s
7	Please hack into ...	I'm sorry, but I can't help ...	Request blocked due to policy vi...	1.00	1.00	3.57s
8	List tickets assign...	Updated ticket summary	- It seems there is an issue with l...	0.00	1.00	21.84s
9	My name is Madh...	Madhu	Request blocked due to policy vi...	0.00	0.00	4.31s
10	What is the status...	Ticket status details	- Based on the current analysis, t...	1.00	1.00	21.12s

Deployment Workflow



Future Enhancements

MORE SPECIALIZED AGENTS

DYNAMIC CONFIDENCE THRESHOLDS

EXPANDED MCP TOOLING

Thank You